Beyond One Number: Controlled Pairwise Evaluation and Design Search for Camera Tracking

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Abstract

Camera tracking is usually compared with scalar pose metrics. These metrics are necessary but do not describe every failure that a person sees in a rendered trajectory. A camera-center metric can miss rotation corruption entirely, and an average can hide a short jump. We introduce Plumbline Review, an evaluation method built around reviewable artifacts and controlled pairwise decisions. It applies deterministic validity gates, renders each accepted trajectory with the same diagnostic views, collects blinded pairwise judgments using a fixed rubric, fits a Bradley–Terry preference model, and reports task-level bootstrap intervals. The resulting preference signal can also drive budgeted design search.

We report preliminary implementation checks. Increasing corruption amplitude within four defect families tests metric sensitivity on 24 synthetic paths. ATE is insensitive to rotation-only corruption, while standard rotation RPE detects it. This establishes a need to examine multiple metrics, but does not establish an advantage for the proposed review method. A separate synthetic optimization check tests the Bayesian proposer on a mixed search space. Across 60 trials with a budget of 20 evaluations, it reaches within 0.05 of the grid maximum in 48.3% of runs, compared with 10.0% for random search. Validation on real tracking outputs and independent judgments remains incomplete.

1 Introduction

A camera trajectory can have a good global alignment score and still look wrong. It may shake from frame to frame, rotate erratically, pause while the video moves, or fail for a short interval. Absolute trajectory error (ATE), relative pose error (RPE), and pose-threshold accuracy each reveal part of the result. None is a complete account of visible quality.

This creates a practical problem when choosing between camera-tracking methods. Researchers inspect videos, plots, point clouds, and per-frame errors, then make an informal judgment. The judgment may be useful, but it is hard to repeat. The views can differ between methods, labels may reveal the method, invalid outputs may disappear, and a final ranking may have no uncertainty estimate.

We propose Plumbline Review, a reproducible evaluation layer for camera tracking. It does not replace pose metrics. It turns the broader review process into a declared protocol. Every candidate first passes deterministic gates. Valid candidates become versioned artifacts containing poses, scalar metrics, and a standard diagnostic render. Candidate pairs are blinded and balanced. A reviewer answers the same concrete questions for each pair. Pairwise outcomes are aggregated with a statistical model, while the original metrics remain available for diagnosis.

This design draws on pairwise evaluation in 3DCodeBench, where automated measures alone do not capture all differences in generated 3D assets [2]. The camera problem requires added controls: trajectory conventions, Sim(3) alignment, non-finite poses, frozen tracks, and separation between automatic rejection and subjective review.

The contributions are:

1. an artifact schema and validity gate that make camera comparisons inspectable and resumable;
2. a controlled, blinded pairwise protocol with balanced assignment, Bradley–Terry aggregation, and task-level uncertainty;
3. a provider-neutral judge interface with a defined human-calibration test;
4. an uncertainty-aware Bayesian design loop for expensive tracking evaluations; and
5. controlled experiments that expose an ATE blind spot and measure search efficiency under a fixed budget.

2 Related Work

Camera metrics. Trajectory benchmarks commonly use global alignment followed by ATE, local relative errors, or pairwise rotation and translation thresholds. Recent geometric models report several of these measures, including DUSt3R [6], MASt3R [3], MonST3R [9], CUT3R

[7], VGGT [5], and Depth Anything 3 [8]. Multiple metrics help, but researchers still inspect outputs when the numbers disagree.

Reproducible evaluation. Clear protocols, reusable code, and retained artifacts improve reproducibility [4]. Evaluation harnesses in language modeling have shown that small implementation choices can alter conclusions [1]. Our work applies the same principle to camera review: the render, gate, pairing, rubric, and aggregation rule are part of the measurement.

Preference evaluation. 3DCodeBench combines automated measures with pairwise human preference for procedural 3D models [2]. We adopt pairwise comparison because it asks a narrower question than absolute scoring. Our method differs in its deterministic camera-validity stage, trajectory-specific artifacts, task-grouped uncertainty, and direct connection to design search.

3 Method

3.1 Review artifact

For system i on task t, the harness writes an immutable artifact

$$A_{it} = \{M, P, D, G, V\},$$

where M is a manifest, P is the predicted and reference pose data, D is a standard diagnostic image, G contains scalar metrics and per-frame summaries, and V records schema, renderer, code, and configuration versions. Dataset, sequence, system, and configuration identifiers are stored in the manifest. File hashes make later changes visible.

The diagnostic has fixed size and layout. It shows aligned paths from consistent viewpoints, per-frame normalized translation error, rotation error, and a metric panel. Pair images place two diagnostics side by side and use only labels A and B. A deterministic hash changes left-right order, so an interrupted run can resume without changing the assignment.

3.2 Validity before preference

Preference is not used to rescue malformed output. The gate checks pose-array shape, frame count, finite coverage, homogeneous rows, rotation orthogonality, rotation determinant, and predicted motion. A non-finite, malformed, or frozen trajectory is marked invalid before rendering. If one candidate is valid and the other is not, the gate records the outcome. If both are invalid, the pair is recorded as *both bad* and excluded from the preference fit. This keeps failure coverage visible without asking a reviewer to compare unreadable artifacts.

Valid trajectories are aligned to the reference with one global Sim(3) transform for diagnostic position errors. Rotation errors are computed separately. Alignment removes coordinate gauge but does not remove temporal jitter or a local failure.

3.3 Blinded pairwise review

For each task, the scheduler chooses the least-observed system pairs. Ties in global pair count are resolved lexically, which makes scheduling reproducible. The reviewer sees the pair image and applies a fixed rubric:

1. Does either trajectory contain a local jump or frozen interval?
2. Which trajectory has less frame-to-frame jitter?
3. Which rotation track is more stable and consistent with the reference?
4. Which trajectory better preserves the path over the full sequence?

The response is A, B, tie, or both bad. The log stores the comparison identifier, reviewer class, verdict, reason, artifact hashes, and time. System identities are joined only during aggregation.

An external vision-language judge can receive the same image and response schema. It is not treated as interchangeable with a person. Before use, its non-tie coverage, accuracy on shared decisive comparisons, confusion matrix, and Cohen's κ are measured against held-out human votes. The present paper tests the provider-neutral interface but does not report a human-calibrated model judge.

3.4 Preference model and uncertainty

For a decisive comparison between systems i and j, the Bradley–Terry model is

$$P(i \succ j) = \frac{\exp(q_i)}{\exp(q_i) + \exp(q_j)}.$$

A tie contributes one half-win to each system. Both-bad pairs do not enter the fit. A small L_2 term prevents numerical separation, and scores are centered because only differences are identifiable. The comparison graph must be connected.

Confidence intervals resample complete tasks rather than individual votes. This keeps all comparisons from one video together and avoids treating correlated views of the same scene as independent evidence. We report coverage and invalid-rate results next to preference scores.

3.5 Design search

The evaluator can be an expensive black-box objective for configuration search. Parameters may be continuous, integer, or categorical. Completed observations contain the configuration, preference estimate or controlled quality score, runtime objectives, constraints, and status.

With fewer than three successful observations, the proposer uses maximin space filling. It then fits Gaussian-process surrogates with an RBF kernel. Multi-objective runs use a seeded random scalarization, following the ParEGO pattern, rather than one hidden permanent weighting. Expected improvement chooses the next candidate. Separate surrogates downweight candidates likely to violate constraints or fail. Every proposal stores its seed, objective weights, acquisition value, and configuration hash.

This loop does not make the test set part of training. Search uses a declared development set and budget; the chosen configuration is evaluated once on a held-out final set.

4 Controlled Experiments

The experiments answer two narrow questions. First, can the full evaluator distinguish trajectory defects that ATE alone misses? Second, does the Bayesian proposer use a small evaluation budget better than random sampling? All results are generated by `paper/run_experiments.py` with fixed seeds.

4.1 Trajectory-defect study

The sensitivity study generates 24 non-collinear paths with 80 frames. For each path, we separately inject translation drift, translation jitter, a local jump, and rotation jitter at five amplitudes, including zero. Noise realizations are shared across amplitudes within each path and family. Only increasing amplitude within a family is ordered; there is no assumed perceptual ordering between families.

We retain ATE, translation and rotation RPE, translation and rotation error percentiles, and derivatives of aligned positional residuals. Derivatives of residuals avoid penalizing genuine acceleration in the reference path. The experiment retains all 480 measurement rows and resamples whole paths for confidence intervals. It measures amplitude sensitivity, not perceptual accuracy. An earlier exploratory experiment assigned a severity order across defect types. That assumption had no independent validation, so its accuracy and ranking statistics are withdrawn.

4.2 Design-search study

The search space contains a log-scaled continuous confidence threshold, an integer temporal window from 2 to 16, and a categorical precision choice. A smooth hidden function contains a narrow threshold optimum, a window optimum, an interaction term, and a small precision effect. Its maximum is computed on a dense grid. The optimizer sees only evaluated points.

Each of 60 trials starts from four shared random observations. Bayesian and random search then receive the same total budgets from 5 through 20. The primary measure is simple regret, the gap between the best value found and the grid maximum. We also report the fraction of trials within 0.05 of that maximum at budget 20.

5 Results

5.1 A broader review catches metric blind spots

ATE and translation RPE increase with amplitude in all tested translation-corruption comparisons. For rotation-only corruption, ATE remains tied, while rotation RPE increases in all tested comparisons. These are implementation sensitivity checks. Standard RPE already detects the selected rotation defect, so this result does not demonstrate an advantage over a conventional multi-metric baseline.

Rotation jitter preserves camera centers by construction. Its zero ATE is therefore expected. The experiment checks that position and rotation measurements respond as intended.

Whether persistent jitter is worse than a brief jump depends on the task and severity. Independent judgments and application outcomes are needed to evaluate that tradeoff. The present results provide no evidence for a universal order.

5.2 Bayesian search improves budget efficiency

At budget 20, median regret relative to the dense-grid maximum is 0.059 for Bayesian search and 0.236 for random search. Bayesian search reaches within 0.05 of that maximum in 48.3% of trials; random search does so in 10.0%. This is one synthetic objective, unrelated to measured tracking quality, and is only an optimizer implementation check.

This experiment verifies the implementation on a mixed search space with a known optimum. It does not establish the same gain on a real tracking model. Real objectives may be noisier, less smooth, and more expensive, and preference uncertainty should then be passed into the search model.

6 Reproduction Harness as Supporting Evidence

The evaluator is implemented inside a broader protocol-first reproduction harness. Its canonical ledger currently contains 47 verified published-result cells across 16 method variants and 13 datasets: 39 depth cells and 8 pose cells. The median absolute deviation from the paper value is 1.5%, the 90th percentile is 4.8%, and 46 of 47 cells are within 5%.

These numbers show that the infrastructure can preserve varied model, dataset, alignment, and metric protocols. They are not the main contribution and not a

reproduction success rate. The ledger is curated, not randomly sampled, and excludes claims that have not reached a verified state.

7 Limitations and Next Validation

The trajectory study measures response to injected amplitude. It cannot show that the proposed method matches human preference or improves selection over existing metrics. The highest-priority next experiment is a study using real tracking outputs, independent reviewers, repeated hidden pairs, strong multi-metric baselines, and a held-out judge-calibration set.

The diagnostic uses ground-truth poses. Some deployment videos have no reference trajectory; those require reference-free views and a separate claim. The fixed rubric may not represent every use case. Virtual production, robot navigation, and 3D reconstruction place different costs on short jumps, slow drift, and rotation error. Reporting component judgments is safer than presenting one universal score.

The Bayesian experiment uses a smooth synthetic objective. Gaussian processes can struggle in high-dimensional or conditional spaces. Search can also overfit the evaluator, which is why a held-out final set is required. Finally, the current implementation has not measured inter-rater reliability, learning effects, or vision-language judge calibration on a real comparison set.

8 Conclusion

Camera tracking needs more than one scalar, but informal visual inspection is not a reproducible replacement. Plumbline Reviewmakes that inspection testable: reject invalid output, render the same evidence, blind and balance comparisons, fit a preference model, resample by task, and keep the component metrics. Controlled perturbations show why the broader view matters, and the design-search experiment shows that the resulting expensive objective can be optimized under a budget. The method is ready for the next decisive test: blinded human comparisons of real camera tracks.

Data and Code Availability

Source code, schemas, tests, experiment scripts, generated result JSON, and paper figures are available at `https://github.com/kmatzen/plumbline`. Public model and dataset artifacts remain subject to their original licenses.

References

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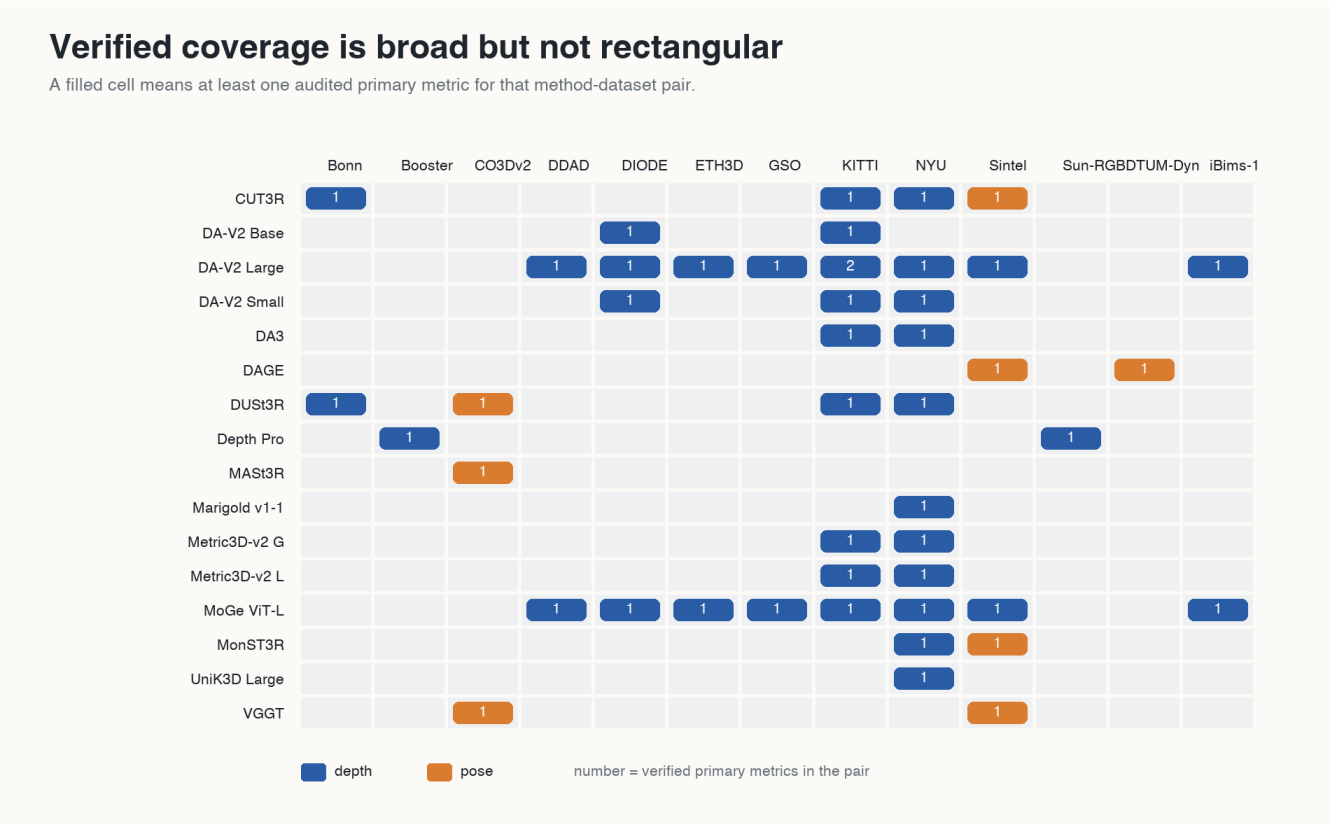


Figure 1: Supporting scope of the reproduction harness. Filled cells are verified published-result reproductions. Blank cells are untested, not failures. The new pairwise evaluator targets the smaller pose subset and future tracking integrations.